

Initial Technical Design

- Desktop application
- Object Oriented Design
 - Test Driven Development
- Python
 - Matplotlib
 - GUI framework

User Personas

Persona: OE Researcher(s)

- Masters student
- Not technically illiterate but not much coding experience
- Wants a system that is simple and quick to use
- Does not like spreadsheets

Description: Collects data and processes the data via the user interface needs the system to be easy to learn because they may not be using it for a long time.

Persona: Administrator (Research Lead, Julie)

- Professor
- Technically experience but more of a hacker than developer
- Wants a system that can be expanded upon

Description: Handles the system and updates it with new modules, needs the system to be updateable/upgradeable because the cast of researchers is changing.

User Stories

*** *means high priority*

** *means medium priority*

* *means low priority*

Nothing means not part of MVP

**As an OE Researcher, I want to be able to export my data to a common format so that I can share it with other researchers.

***As an OE Researcher I want to be able to select different data processing methods within the same application so that I do not need to use multiple systems to process the same data.

***As an OE Researcher, I want a simple unified code base so that I can focus on analytics rather than data pipelines.

*** As an OE Researcher, I want a single interface that is easy to interact with to analyze penetrometer data.

***As an OE Researcher, I want to be able to generate a visual of the data so that I can properly analyze my results.

**As an OE Researcher, I want to be able to account for uncertainty within my data so that I can be more confident in my analysis.

**As an OE Researcher, I want to be able to generate a condensed input file with only the data around the actual drop points, so there is no useless data included.

*As an Administrator, I want to be able to easily add more data processing modules so that this tool can be updated in the future.

**As an Administrator, I want to be able to easily edit data processing modules so that this tool can be adapted in the future.

As an Administrator, I want to be able to view a history of data input/manipulation so that I can see how the system has evolved over time.

*As an OE Researcher, I want to be able to import previously processed data, so I can view someone else's results.

**As an OE Researcher, I want to be able to input save constants for my penetrometer, so that I don't need to enter them every time I use the program.

**As an OE Researcher, I want to be able to enter constants for my penetrometer so that the equations can handle the penetrometer's specific configuration.

**As an OE Researcher, I want to be able to utilize previous penetrometer configurations so that I can save time.

*As an OE Researcher, I want to be able to view multiple drop points so at once I can compare them directly.

*As an OE Researcher, I want to be able to view each integration/intermediary step of deceleration, velocity, and bearing output of soil, so that I can better understand the data.

Expanded Stories

- Story: As an OE Researcher, I want a simple unified code base so that I can focus on analytics rather than data pipelines.
- Definition of Done
 - Unify existing data analysis code
 - Be able to analyze raw data without moving programs
 - Ability to visualize data
- Tasks
 - Create one application for data analysis
- Steps
 - Research how to make application
 - Analyze existing analysis scripts
 - Rewrite existing analysis scripts into one unified system

- Story: As an OE Researcher, I want to be able to generate a visual of the data so that I can properly analyze my results.
- Definition of Done
 - After inputting penetrometer data into the system an OE Researcher can use the generated visuals to determine the type of soil
 - A graph that plots acceleration over time is generated
 - There is a way to select the specific data an OE Researcher wants to visualize
- Tasks
 - Frontend a way to select the desired data
 - Frontend a way to display the selected data
 - Backend combining the existing systems to use the OE Researches rules and constants to generate effective visuals
- Steps
 - Research a graphical display library
 - Look into the existing tools to see what can be expanded and what needs to be rewritten
 - Integrate graphical display into desktop application